

AWS CloudWatch Lab: Create and Test an EC2 CPU Utilization Alarm

Summary:-

In this lab, an **Amazon EC2** instance was launched with a User Data script that created a web page to generate CPU load. An **Amazon CloudWatch Alarm** was configured to monitor the EC2 instance's CPU utilization and trigger when the average CPU usage exceeded a defined threshold. An **Amazon SNS** topic was created to send email notifications whenever the alarm entered the **In Alarm** state. Finally, the alarm was tested by increasing the CPU load, verifying the alarm status from both the CloudWatch and EC2 consoles, receiving the email notification, and cleaning up all created resources.

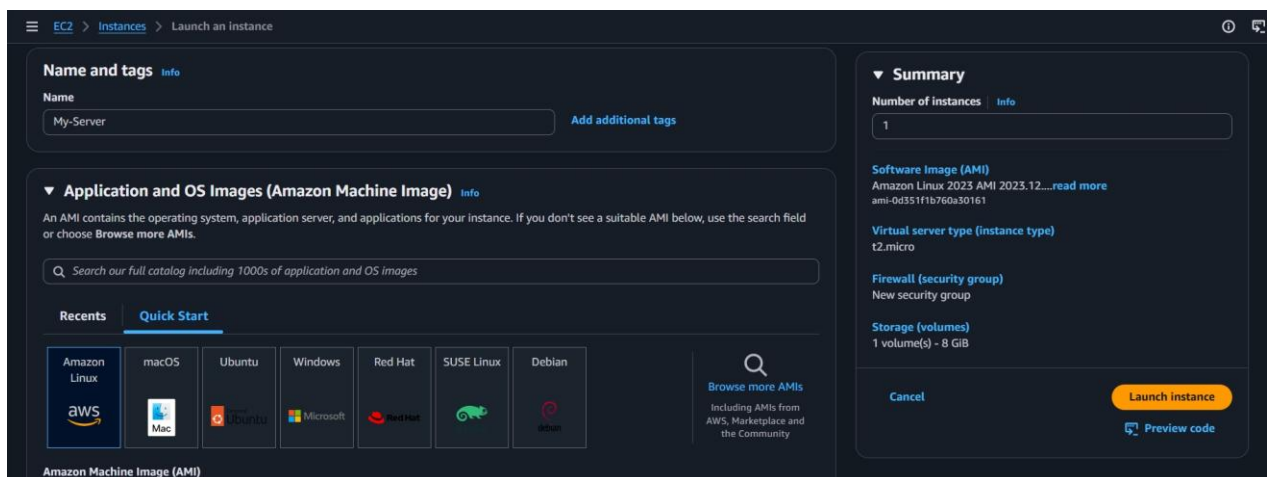
Let's Start:-

- Ensure an EC2 instance is available before creating a CloudWatch alarm.
- The User Data script creates a web page that can increase or decrease CPU utilization for testing.
- Port **80 (HTTP)** must be allowed in the security group to access the web page.

Lab Steps:-

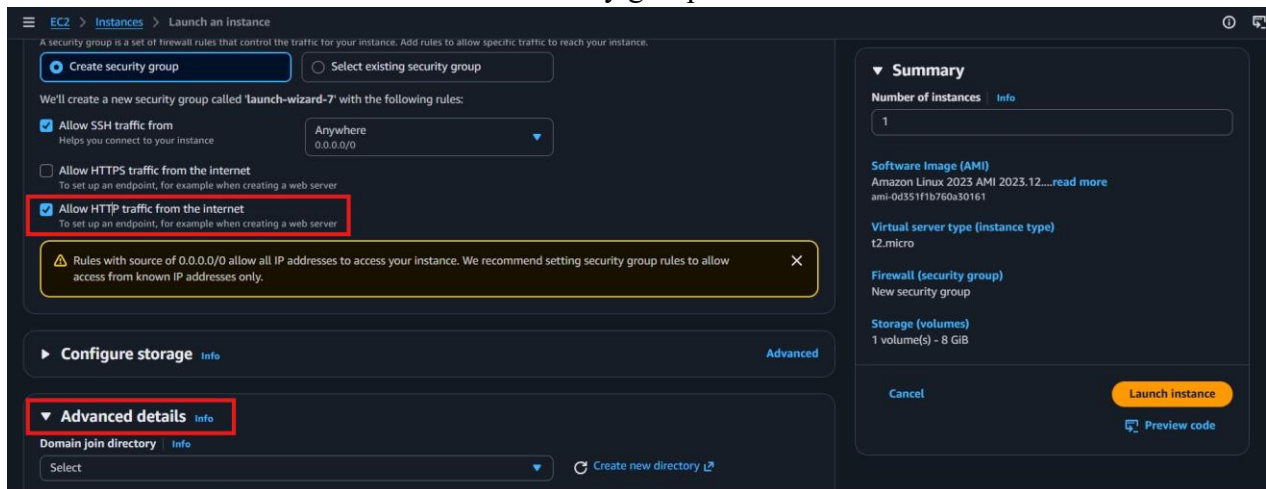
Step 1 – Launch an EC2 Instance

- Sign in to the AWS Management Console.
- Navigate to **Amazon EC2**.
- Click **Launch Instance**.
- Enter the instance name (e.g., **My Server**).
- Select **Amazon Linux** as the AMI.



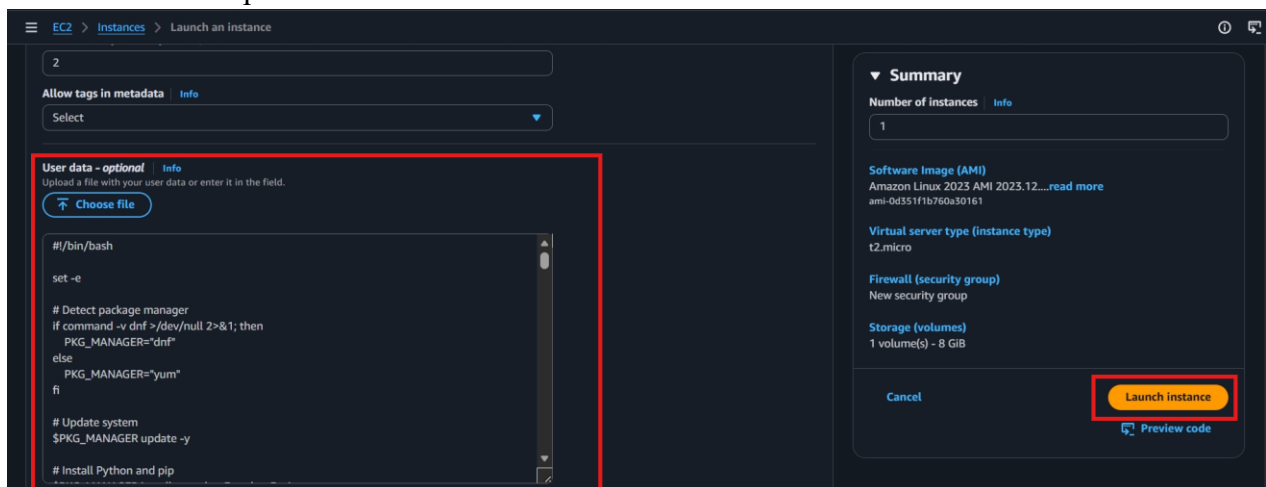
- Choose a Free Tier eligible instance type (t2.micro or t3.micro).
- Select an existing Key Pair.

- Use the Default VPC.
- Enable Allow HTTP Traffic in the security group.



Step 2 – Add the User Data Script

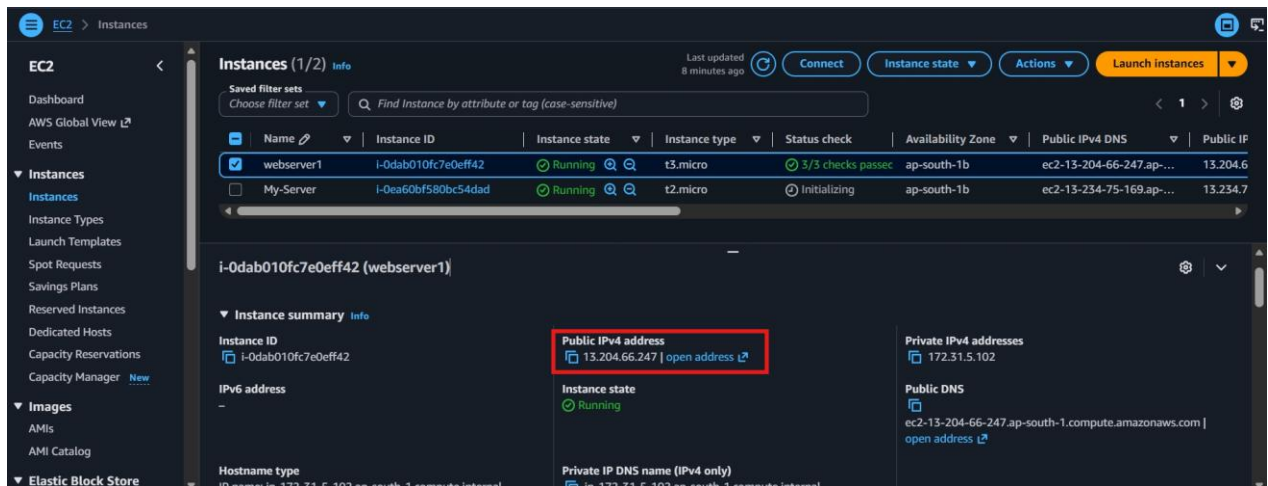
- Scroll to Advanced Details.
- Open the provided User Data Script (status.sh) file from the lab resources.
- Copy the complete script.
- Paste the script into the User Data section.



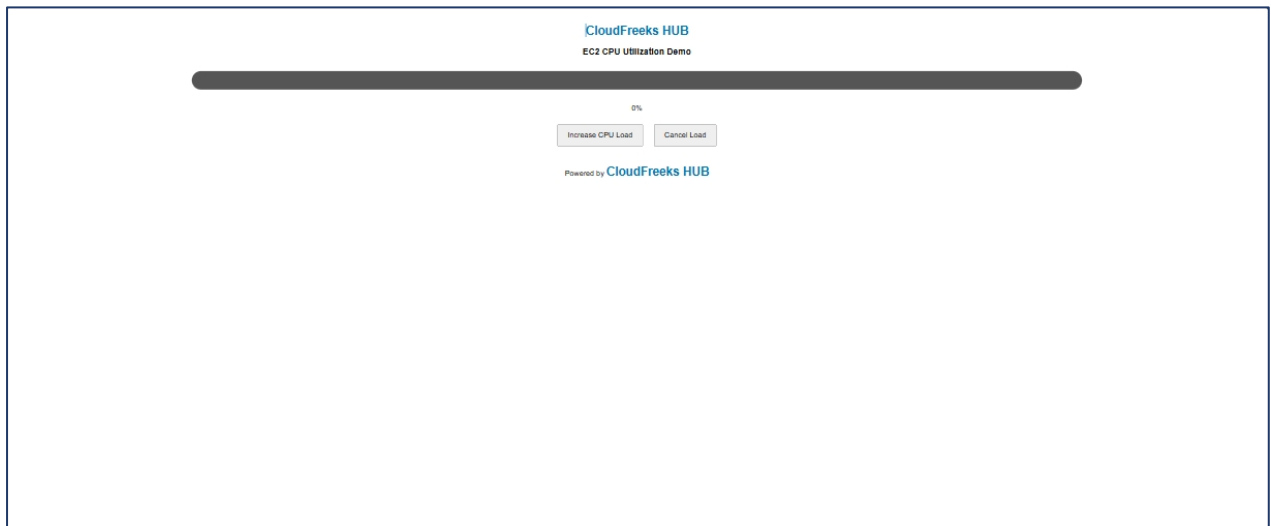
- Click Launch Instance.
- Wait until the instance status changes to Running and Status Checks pass successfully.

Step 3 – Verify the Web Application

- Copy the Public IPv4 Address of the EC2 instance.



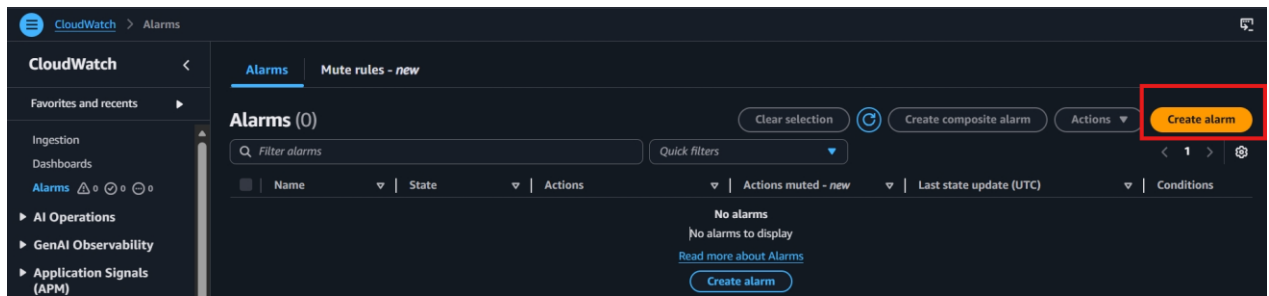
- Open a web browser.
- Access the application using:
http://<Public-IP>
- Verify that the web page opens successfully.



- Do not increase the CPU load yet.

Step 4 – Create a CloudWatch Alarm

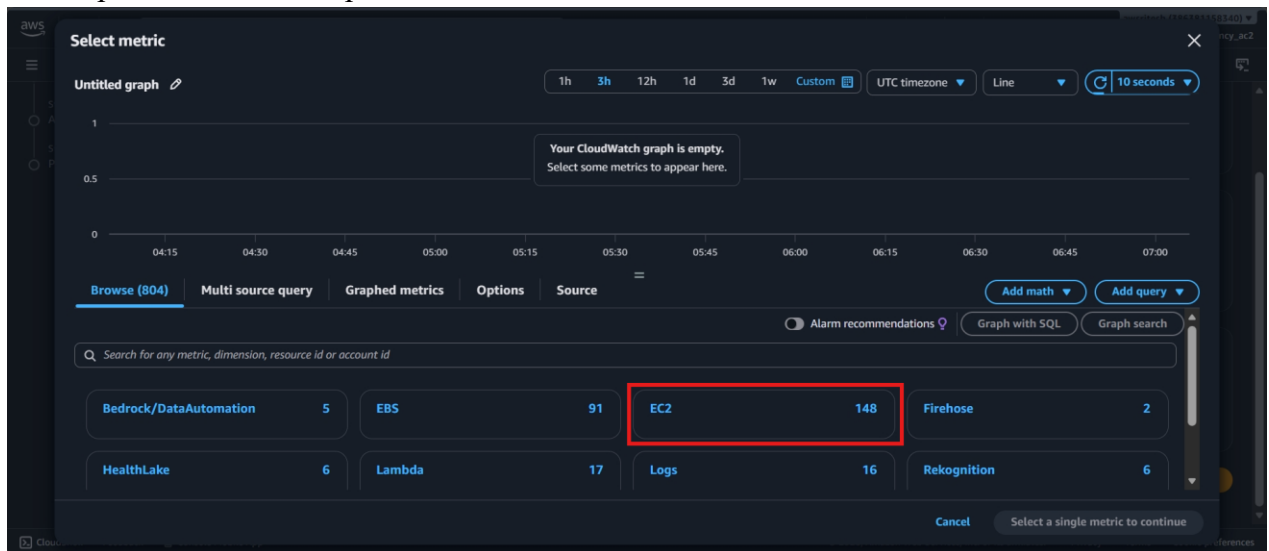
- Navigate to Amazon CloudWatch.
- Select Alarms from the left navigation pane.
- Click Create Alarm.



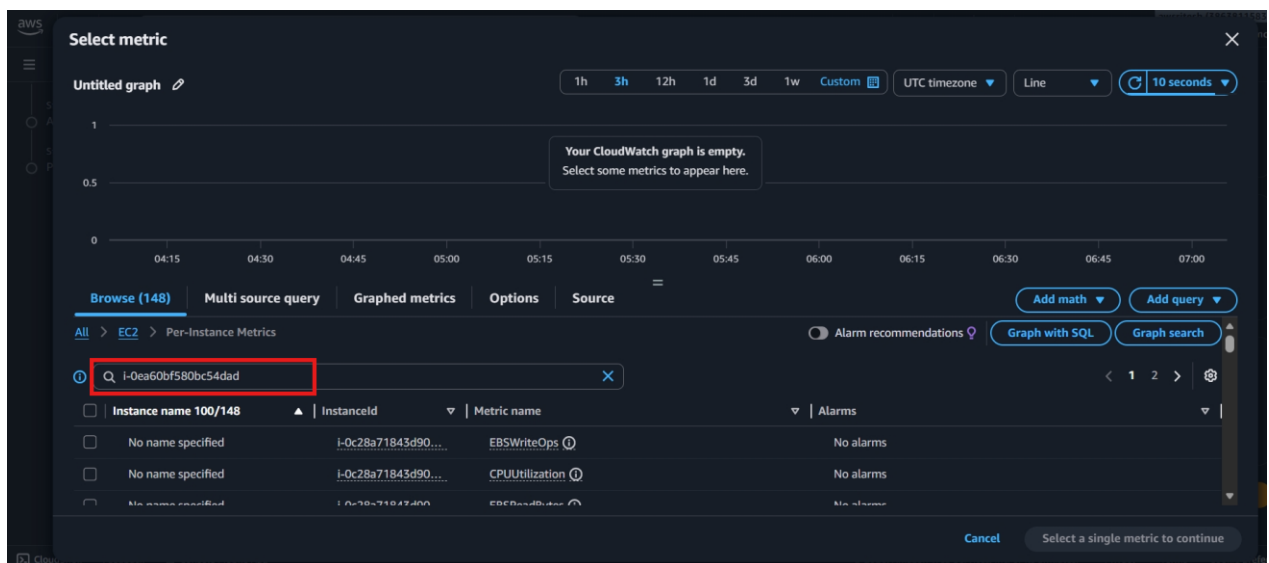
- Choose Classic Mode.
- Click Select Metric.

Step 5 – Select the CPU Utilization Metric

- Open the EC2 namespace.



- Select Per-Instance Metrics.
- Search for the EC2 Instance ID.



- Select the CPUUtilization metric.
- Click Select Metric.

The screenshot shows the 'Browse' page in AWS CloudWatch. It displays a table of metrics for the resource 'My-Server'. The metrics listed are: StatusCheckFailed_System, NetworkPacketsIn, NetworkOut, NetworkIn, NetworkPacketsOut, and CPUUtilization. The 'CPUUtilization' metric is selected, indicated by a blue checkmark in the first column. At the bottom right, there are 'Cancel' and 'Select metric' buttons. The 'Select metric' button is highlighted with a red box.

Step 6 – Configure the Alarm Conditions

- Keep the Statistic as Average.
- Keep the Period as 5 Minutes.
- Select Static Threshold.
- Set the condition to Greater than 80%.
- Keep the evaluation as 1 Data Point.
- Click Next.

The screenshot shows the 'Create alarm' page in AWS CloudWatch. The 'Whenever CPUUtilization is...' section is configured with the 'Greater' condition and a threshold of 80. The 'than...' section shows the threshold value '80'. The 'Additional configuration' section shows 'Datapoints to alarm' set to '1' out of '1'. The 'Missing data treatment' is set to 'Treat missing data as missing'. At the bottom right, there are 'Cancel' and 'Next' buttons. The 'Next' button is highlighted with a red box.

Step 7 – Configure Email Notifications

- Under Notification, select Create New Topic.
- Enter a topic name or use the default name.
- Enter your email address.

CloudWatch > Alarms > Create alarm

Step 3
Add alarm details

Step 4
Preview and create

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic

☒ **Create new topic**

☐ Use topic ARN to notify other accounts

Create a new topic...
The topic name must be unique.

Default_CloudWatch_Alarms_Topic

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

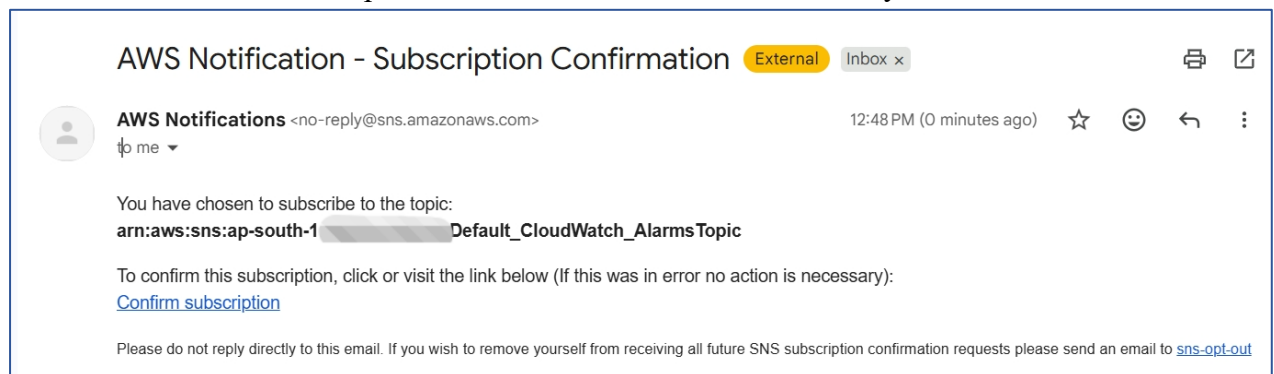
mytesting@gmail.com

user1@example.com, user2@example.com

Create topic

Add notification

- Click Create Topic.
- Open your email inbox.
- Click Confirm Subscription from the confirmation email sent by AWS.



- Return to the CloudWatch console.

Step 8 – Create the Alarm

- Enter an alarm description.
- Enter an alarm name (e.g., My Server-Alarm).
- Click Create Alarm.

CloudWatch > Alarms > Create alarm

Step 1: Specify metric and conditions
 Step 2: Configure actions
 Step 3: **Add alarm details**
 Step 4: Preview and create

Add alarm details

Name and description

Alarm name: MyServer-Alarm_

Alarm description - optional [View formatting guidelines](#)

Edit Preview

This is Test Alarm

Up to 1024 characters (18/1024).

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional [Info](#)

No tags associated with the resource.

[Add new tag](#)

- Verify that the alarm is created successfully.

CloudWatch > Alarms > Create alarm

Step 3: Add alarm details

Actions

Notification: When In alarm, send a notification to "Default_CloudWatch_AlarmsTopic"

Alarm details

Name: MyServer-Alarm_
 Description: This is Test Alarm

Tags (0)

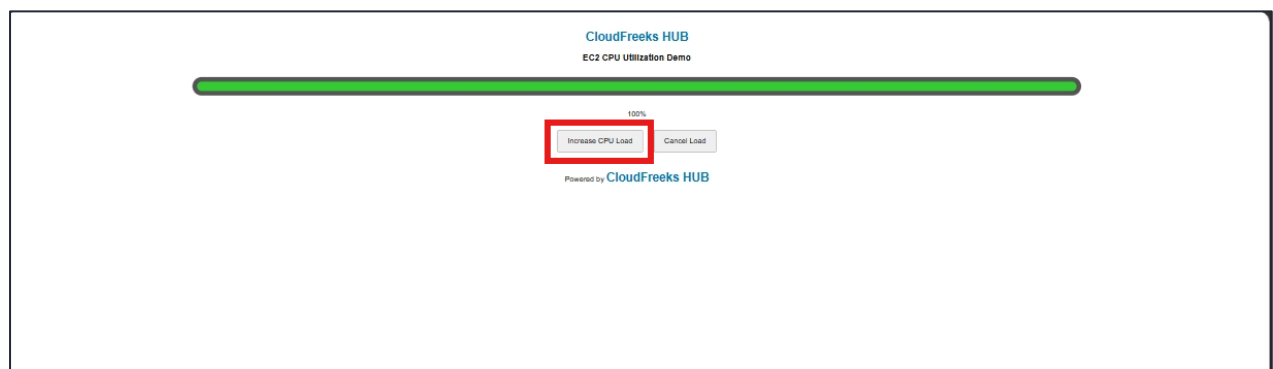
Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Cancel Previous **Create alarm**

- Initially, the alarm status may display Insufficient Data until CloudWatch collects metrics.

Step 9 – Generate CPU Load

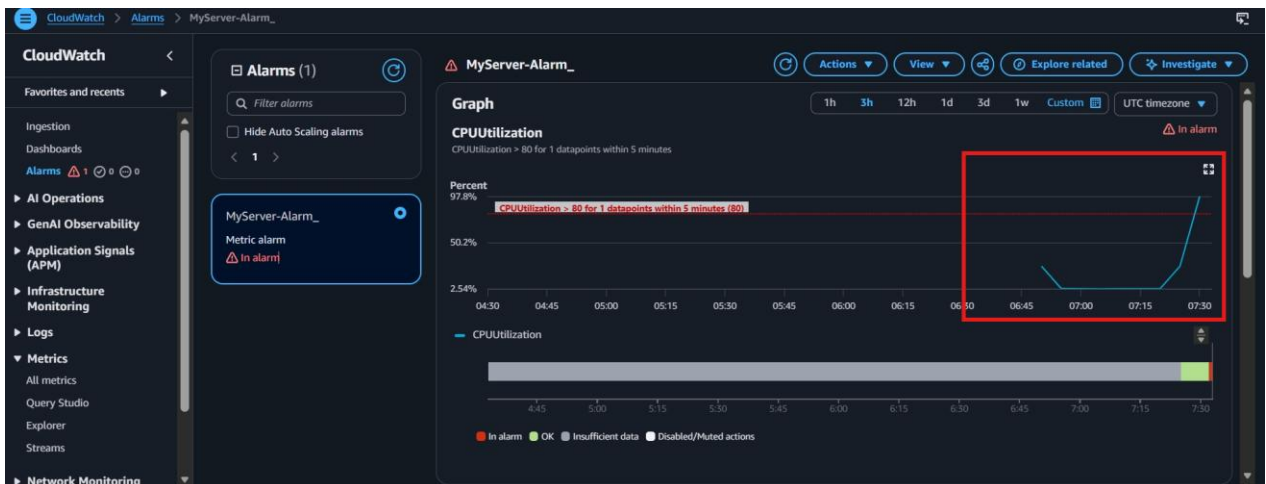
- Return to the EC2 web application.
- Click Increase CPU Load.



- Wait several minutes for the CPU utilization to increase.
- Monitor the alarm status in the CloudWatch console.
- Observe the CPU utilization graph as it approaches the configured threshold.

Step 10 – Verify the Alarm

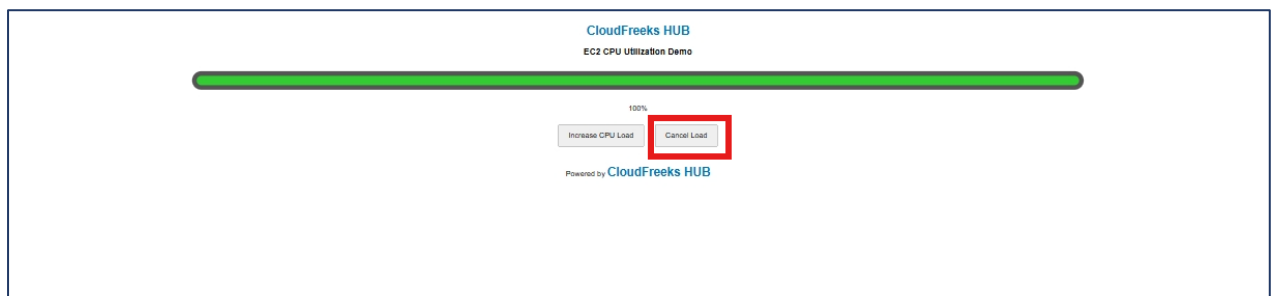
- Confirm the alarm state changes from OK to In Alarm after the average CPU utilization exceeds 80%.



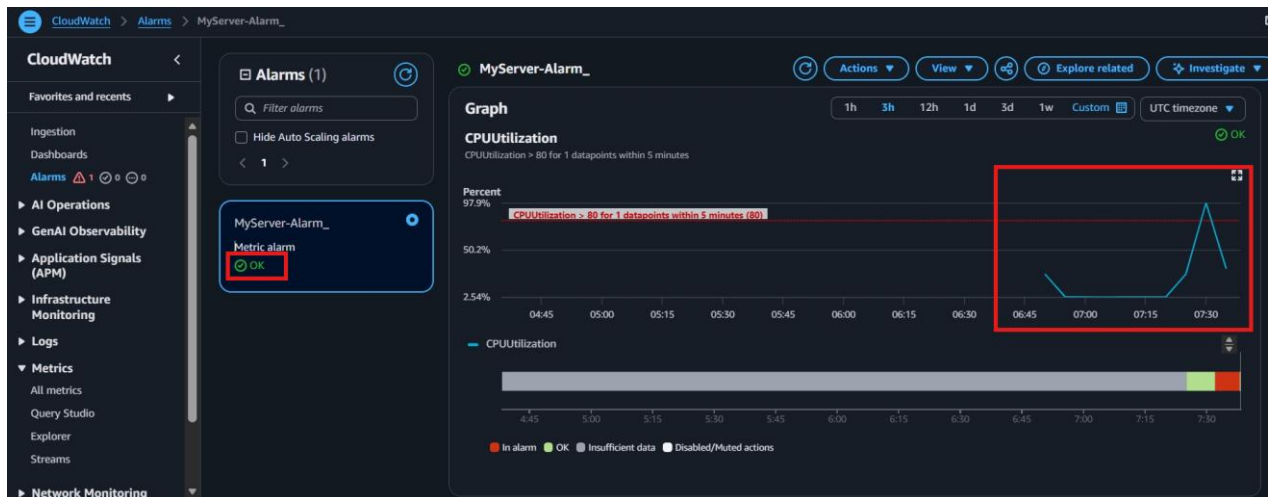
- Verify the alarm status from the CloudWatch console.
- Navigate to the EC2 console and verify the alarm status under Status and Alarms.
- Check your email inbox and confirm that an Amazon SNS notification has been received.

Step 11 – Stop the CPU Load

- Return to the web application.
- Click Cancel CPU Load.

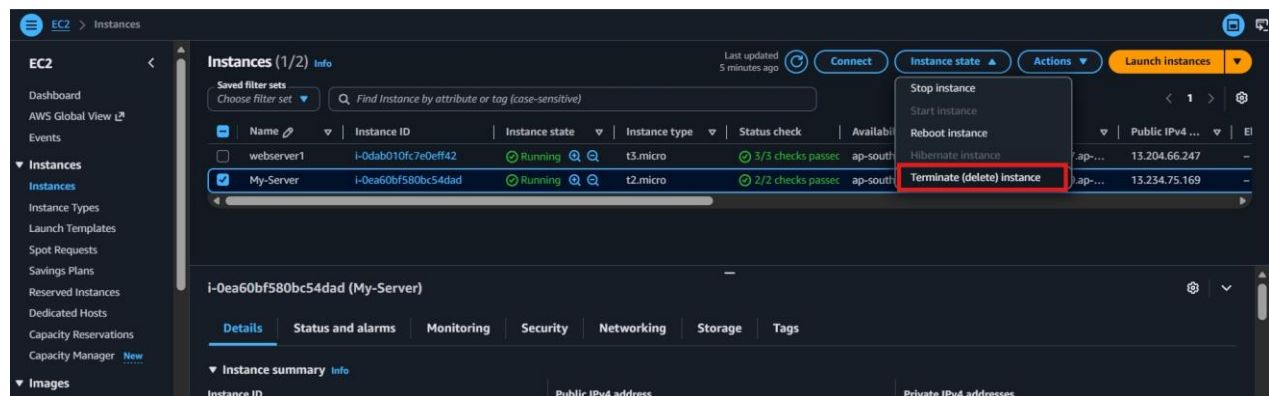


- Wait for the average CPU utilization to decrease.
- Observe that the alarm eventually returns to the OK state.



Step 12 – Clean Up Resources

- Navigate to Amazon CloudWatch.
- Select the created alarm.
- Click Actions → Delete.
- Navigate to Amazon EC2.
- Select the EC2 instance.
- Click Instance State → Terminate Instance.



Verification

- The EC2 instance was launched successfully.
- The User Data script created the CPU load testing web application.
- A CloudWatch alarm was configured to monitor CPUUtilization.
- An Amazon SNS email notification was successfully configured and confirmed.
- The alarm entered the In Alarm state when CPU utilization exceeded 80%.
- The alarm status was verified from both the CloudWatch and EC2 consoles.
- The email notification was received successfully.
- The created alarm and EC2 instance were deleted after completing the lab.